Soft Computing — Unit 2: Fuzzy Sets, Fuzzy Logic & Fuzzy Systems

> **Syllabus:** Classical sets and fuzzy sets, fuzzy relations, operations & properties, membership functions, fuzzification methods, λ-cuts & defuzzification, classical & fuzzy logic, approximate reasoning, fuzzy implication, fuzzy rule based systems (linguistic hedges, aggregation, Mamdani & Sugeno FIS), applications (home appliances, controllers, medical diagnosis, weather forecasting).

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1. Classical Sets — Operations & Properties

📘 1.1 Definition

A **classical (crisp) set** A over a universe of discourse X is a collection where every $x \in X$ either *fully belongs* or *does not belong*:

$$\chi_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

χ_A is called the **characteristic function**.

Representation

- **Roster form:** $A = \{1, 2, 3, 5, 7\}$
- **Set-builder:** $A = \{x \in \mathbb{N} : x \leq 7, x \text{ is prime or } 1\}$

- **Characteristic vector:** for $X = \{1, 2, \ldots, 8\}$, $\chi_A = (1, 1, 1, 0, 1, 0, 1, 0)$

🧮 1.2 Operations on Classical Sets

Let $A, B \subseteq X$.

Operation	Definition	Characteristic Function
Union	$A \cup B = \{x : x \in A \text{ or } x \in B\}$	$\chi_{A\cup B}(x) = \max(\chi_A, \chi_B)$
Intersection	$A \cap B = \{x : x \in A \text{ and } x \in B\}$	$\chi_{A\cap B}(x) = \min(\chi_A, \chi_B)$
Complement	$\bar{A} = \{x \in X : x \notin A\}$	$\chi_{\bar{A}}(x) = 1 - \chi_A(x)$
Difference	$A - B = A \cap \bar{B}$	$\chi_{A-B}(x) = \chi_A \cdot (1 - \chi_B)$
Sym. difference	$A \triangle B = (A - B) \cup (B - A)$	XOR
Cartesian product	$A \times B = \{(a, b) : a \in A, b \in B\}$	—

📐 1.3 Properties of Classical Sets

Law	Statement
Commutative	$A \cup B = B \cup A, A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$
Distributive	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Idempotent	$A \cup A = A, A \cap A = A$
Identity	$A \cup \emptyset = A, A \cap X = A$
Domination	$A \cup X = X, A \cap \emptyset = \emptyset$
Involution	$\bar{\bar{A}} = A$
De Morgan's	$\overline{A \cup B} = \bar{A} \cap \bar{B}$
Absorption	$A \cup (A \cap B) = A, A \cap (A \cup B) = A$
Law of Excluded Middle	$A \cup \bar{A} = X$ ✅ holds
Law of Contradiction	$A \cap \bar{A} = \emptyset$ ✅ holds

> ⚠️ The last two laws hold for crisp sets but **fail** for fuzzy sets — this is *the* defining difference.

Cardinality (crisp)

$$|A| = \sum_{x \in X} \chi_A(x)$$

For a power set: $|2^X| = 2^{|X|}$.

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2. Fuzzy Sets — Operations & Properties

📘 2.1 Definition (Recap)

A **fuzzy set** $\tilde{A}$ over X is defined by a membership function $\mu_{\tilde{A}} : X \to [0,1]$.

$$\tilde{A} = \{(x, \mu_{\tilde{A}}(x)) : x \in X\}$$

Zadeh's notation:

$$\tilde{A} = \sum_{x_i \in X} \frac{\mu_{\tilde{A}}(x_i)}{x_i} \quad \text{(discrete)} \qquad \tilde{A} = \int_X \frac{\mu_{\tilde{A}}(x)}{x} \quad \text{(continuous)}$$

> The "+" and "/" are **symbolic**, *not* arithmetic.

🧮 2.2 Operations on Fuzzy Sets (Zadeh's Standard)

Let $\tilde{A}, \tilde{B}$ be fuzzy sets on X.

Operation	Membership Function
Union	$\mu_{\tilde{A}\cup\tilde{B}}(x) = \max(\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x))$
Intersection	$\mu_{\tilde{A}\cap\tilde{B}}(x) = \min(\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x))$
Complement	$\mu_{\bar{\tilde{A}}}(x) = 1 - \mu_{\tilde{A}}(x)$
Equality	$\tilde{A} = \tilde{B} \iff \mu_{\tilde{A}}(x) = \mu_{\tilde{B}}(x) \forall x$
Containment	$\tilde{A} \subseteq \tilde{B} \iff \mu_{\tilde{A}}(x) \leq \mu_{\tilde{B}}(x) \forall x$
Algebraic product	$\mu_{\tilde{A}\cdot\tilde{B}}(x) = \mu_{\tilde{A}}(x) \cdot \mu_{\tilde{B}}(x)$
Algebraic sum	$\mu_{\tilde{A}+\tilde{B}}(x) = \mu_{\tilde{A}}(x) + \mu_{\tilde{B}}(x) - \mu_{\tilde{A}}(x)\mu_{\tilde{B}}(x)$

Bounded sum	$\mu(x) = \min(1, \mu_{\tilde{A}}(x) + \mu_{\tilde{B}}(x))$
Bounded difference	$\mu(x) = \max(0, \mu_{\tilde{A}}(x) - \mu_{\tilde{B}}(x))$
Power	$\tilde{A}^p$ has $\mu(x) = [\mu_{\tilde{A}}(x)]^p$

Generalized — *t*-norms & *s*-norms (*t*-conorms)

A **t-norm** $T : [0,1]^2 \to [0,1]$ generalizes intersection; an **s-norm** S generalizes union.

Family	**t-norm** $T(a,b)$	**s-norm** $S(a,b)$
Standard (Zadeh)	$\min(a,b)$	$\max(a,b)$
Algebraic	$a \cdot b$	$a + b - ab$
Łukasiewicz	$\max(0, a + b - 1)$	$\min(1, a + b)$
Drastic	a if $b = 1$, b if $a = 1$, else 0	dual

Axioms (t-norm): commutativity, associativity, monotonicity, boundary $T(a,1) = a$.

📐 2.3 Properties of Fuzzy Sets

Property	**Status**
Commutativity	✅
Associativity	✅
Distributivity	✅
Idempotency	✅ $(\max, \min$ semantics)
Identity & Domination	✅
De Morgan's laws	✅
Involution	✅
Law of Excluded Middle $\tilde{A} \cup \bar{\tilde{A}} = X$	❌ FAILS
Law of Contradiction $\tilde{A} \cap \bar{\tilde{A}} = \emptyset$	❌ FAILS

📐 Why excluded middle fails — example

If $\mu_{\tilde{A}}(x) = 0.4$ then $\mu_{\bar{\tilde{A}}}(x) = 0.6$, so
$\mu_{\tilde{A}\cup\bar{\tilde{A}}}(x) = \max(0.4, 0.6) = 0.6 \neq 1$ and
$\mu_{\tilde{A}\cap\bar{\tilde{A}}}(x) = \min(0.4, 0.6) = 0.4 \neq 0$.

🧮 2.4 Cardinality of a Fuzzy Set

Scalar (sigma) cardinality

$$|\tilde{A}| = \sum_{x\in X} \mu_{\tilde{A}}(x) \qquad \text{(discrete)} \qquad |\tilde{A}| = \int_X \mu_{\tilde{A}}(x)dx \qquad \text{(continuous)}$$

Relative cardinality

$$\|\tilde{A}\| = \frac{|\tilde{A}|}{|X|}$$

Fuzzy cardinality

$|\tilde{A}|_f$ is itself a fuzzy set whose elements are α-cut cardinalities at degree α.

✏️ 2.5 Solved Problems

✏️ Problem 1 — Operations & Cardinality

$X = \{a, b, c, d, e\}$,
$\tilde{A} = \frac{0.1}{a} + \frac{0.4}{b} + \frac{0.7}{c} + \frac{1.0}{d} + \frac{0.3}{e}$,
$\tilde{B} = \frac{0.5}{a} + \frac{0.6}{b} + \frac{0.2}{c} + \frac{0.8}{d} + \frac{0.4}{e}$.

Find $\tilde{A}\cup\tilde{B}$, $\tilde{A}\cap\tilde{B}$, $\bar{\tilde{A}}$, $|\tilde{A}|$, $\|\tilde{A}\|$, algebraic product and bounded sum.

x	μ_A	μ_B	$\cup$	$\cap$	$1-\mu_A$	$\mu_A\mu_B$	$\min(1, \mu_A+\mu_B)$
a	0.1	0.5	0.5	0.1	0.9	0.05	0.6
b	0.4	0.6	0.6	0.4	0.6	0.24	1.0
c	0.7	0.2	0.7	0.2	0.3	0.14	0.9
d	1.0	0.8	1.0	0.8	0.0	0.80	1.0
e	0.3	0.4	0.4	0.3	0.7	0.12	0.7

$|\tilde{A}| = 0.1 + 0.4 + 0.7 + 1.0 + 0.3 = \mathbf{2.5}$
$\|\tilde{A}\| = 2.5/5 = \mathbf{0.5}$

✏️ Problem 2 — Verify Excluded-Middle Failure

For $\mu_{\tilde{A}}(x) = 0.3$ on a singleton universe, show $\tilde{A} \cup \bar{\tilde{A}} \neq X$.

$\mu_{\tilde{A} \cup \bar{\tilde{A}}} = \max(0.3, 0.7) = 0.7 \neq 1$. ✅ Demonstrated.

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3. Fuzzy Relations

📘 3.1 Crisp Relation Recap

A crisp relation R from X to Y is a subset of $X \times Y$:

$$\chi_R(x,y) = \begin{cases} 1 & (x,y) \in R \\ 0 & \text{else} \end{cases}$$

A **fuzzy relation** $\tilde{R}$ generalizes this with $\mu_{\tilde{R}} : X \times Y \to [0,1]$.

$$\tilde{R} = \{((x,y), \mu_{\tilde{R}}(x,y)) : x \in X, y \in Y\}$$

Represented as a **fuzzy relation matrix** $M_{\tilde{R}} = [\mu_{\tilde{R}}(x_i, y_j)]$.

🧮 3.2 Operations on Fuzzy Relations

Let $\tilde{R}, \tilde{S}$ be fuzzy relations on $X \times Y$.

Operation	Definition
Union	$\mu_{\tilde{R} \cup \tilde{S}}(x,y) = \max(\mu_{\tilde{R}}, \mu_{\tilde{S}})$
Intersection	$\mu_{\tilde{R} \cap \tilde{S}}(x,y) = \min(\mu_{\tilde{R}}, \mu_{\tilde{S}})$
Complement	$\mu_{\bar{\tilde{R}}}(x,y) = 1 - \mu_{\tilde{R}}(x,y)$
Containment	$\tilde{R} \subseteq \tilde{S} \iff \mu_{\tilde{R}} \leq \mu_{\tilde{S}}$
Inverse / Transpose	$\mu_{\tilde{R}^{-1}}(y,x) = \mu_{\tilde{R}}(x,y)$
Projection on X	$\mu_{\text{proj}_X \tilde{R}}(x) = \max_y \mu_{\tilde{R}}(x,y)$
Cylindric extension	from $X \to X \times Y$ by repeating μ over Y

🧮 3.3 Composition of Fuzzy Relations

Given $\tilde{R}$ on $X \times Y$ and $\tilde{S}$ on $Y \times Z$, the composition $\tilde{R} \circ \tilde{S}$ on $X \times Z$:

Max–Min Composition

$$\mu_{\tilde{R}\circ\tilde{S}}(x,z) = \max_{y\in Y} \min\big(\mu_{\tilde{R}}(x,y), \mu_{\tilde{S}}(y,z)\big)$$

Max–Product Composition

$$\mu_{\tilde{R}\circ\tilde{S}}(x,z) = \max_{y\in Y}\big(\mu_{\tilde{R}}(x,y)\cdot\mu_{\tilde{S}}(y,z)\big)$$

> Composition is analogous to matrix multiplication where $(+,\times)\to(\max,\min)$ or $(\max,\cdot)$.

📐 3.4 Properties of Fuzzy Relations on $X \times X$

A fuzzy relation $\tilde{R}$ on $X \times X$ may have:

Property	Definition
Reflexive	$\mu_{\tilde{R}}(x,x) = 1 \forall x$
Symmetric	$\mu_{\tilde{R}}(x,y) = \mu_{\tilde{R}}(y,x)$
Anti-symmetric	If $\mu_{\tilde{R}}(x,y) > 0$ and $\mu_{\tilde{R}}(y,x) > 0$ then $x = y$
Transitive (max-min)	$\mu_{\tilde{R}}(x,z) \geq \max_y \min(\mu_{\tilde{R}}(x,y), \mu_{\tilde{R}}(y,z))$
Fuzzy equivalence	Reflexive + Symmetric + Transitive
Fuzzy tolerance	Reflexive + Symmetric

Cardinality of a Fuzzy Relation

$$|\tilde{R}| = \sum_{x,y} \mu_{\tilde{R}}(x,y)$$

✏️ 3.5 Solved Problem — Max-Min Composition

$X = \{x_1, x_2\}, Y = \{y_1, y_2, y_3\}, Z = \{z_1, z_2\}$.

$$\tilde{R} = \begin{bmatrix} 0.3 & 0.8 & 0.5 \\ 0.7 & 0.2 & 0.9 \end{bmatrix} \qquad \tilde{S} = \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \\ 0.5 & 0.8 \end{bmatrix}$$

Compute $\tilde{T} = \tilde{R} \circ \tilde{S}$ via max-min.

Compute $\mu_{\tilde{T}}(x_1, z_1)$:

$$\max\{\min(0.3,0.6), \min(0.8,0.3), \min(0.5,0.5)\} = \max\{0.3,0.3,0.5\} = 0.5$$

$\mu_{\tilde{T}}(x_1, z_2)$: $\max\{\min(0.3, 0.4), \min(0.8, 0.7), \min(0.5, 0.8)\} = \max\{0.3, 0.7, 0.5\} = 0.7$

$\mu_{\tilde{T}}(x_2, z_1)$: $\max\{\min(0.7, 0.6), \min(0.2, 0.3), \min(0.9, 0.5)\} = \max\{0.6, 0.2, 0.5\} = 0.6$

$\mu_{\tilde{T}}(x_2, z_2)$: $\max\{\min(0.7, 0.4), \min(0.2, 0.7), \min(0.9, 0.8)\} = \max\{0.4, 0.2, 0.8\} = 0.8$

$$\tilde{T} = \begin{bmatrix} 0.5 & 0.7 \\ 0.6 & 0.8 \end{bmatrix}$$

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4. Membership Functions & Fuzzification Methods

📘 4.1 Features of an MF

A membership function $\mu_{\tilde{A}}(x)$ is characterized by:

Feature	Meaning
Core	$\{x : \mu_{\tilde{A}}(x) = 1\}$ — full membership
Support	$\{x : \mu_{\tilde{A}}(x) > 0\}$ — non-zero region
Boundary	$\{x : 0 < \mu_{\tilde{A}}(x) < 1\}$ — partial membership
Height	$\sup_x \mu_{\tilde{A}}(x)$
Normal	Height = 1
Sub-normal	Height < 1
Crossover point	$\mu(x) = 0.5$
Symmetry	$\mu(c + t) = \mu(c - t)$ around center c
Convex	$\mu(\lambda x_1 + (1 - \lambda)x_2) \geq \min(\mu(x_1), \mu(x_2))$

📐 Convexity Theorem

A fuzzy set $\tilde{A}$ is convex **iff** every α-cut A_α is a convex (interval) crisp set.

```
  1 ┤          ┌──────┐                    ← Core (μ=1)
    │        /│      │\
  μ │       / │      │ \
    │      /  │      │  \     Boundary
  0 ┤────/───┴──────┴───\─────
           └─── Support ───┘
             Symmetric, Normal, Convex MF
```

🧮 4.2 Standard MF Forms

A. Triangular MF

$$\mu(x;a,b,c) = \max\left(\min\left(\frac{x-a}{b-a}, \frac{c-x}{c-b}\right), 0\right)$$

Parameters: a = left foot, b = peak (μ = 1), c = right foot.

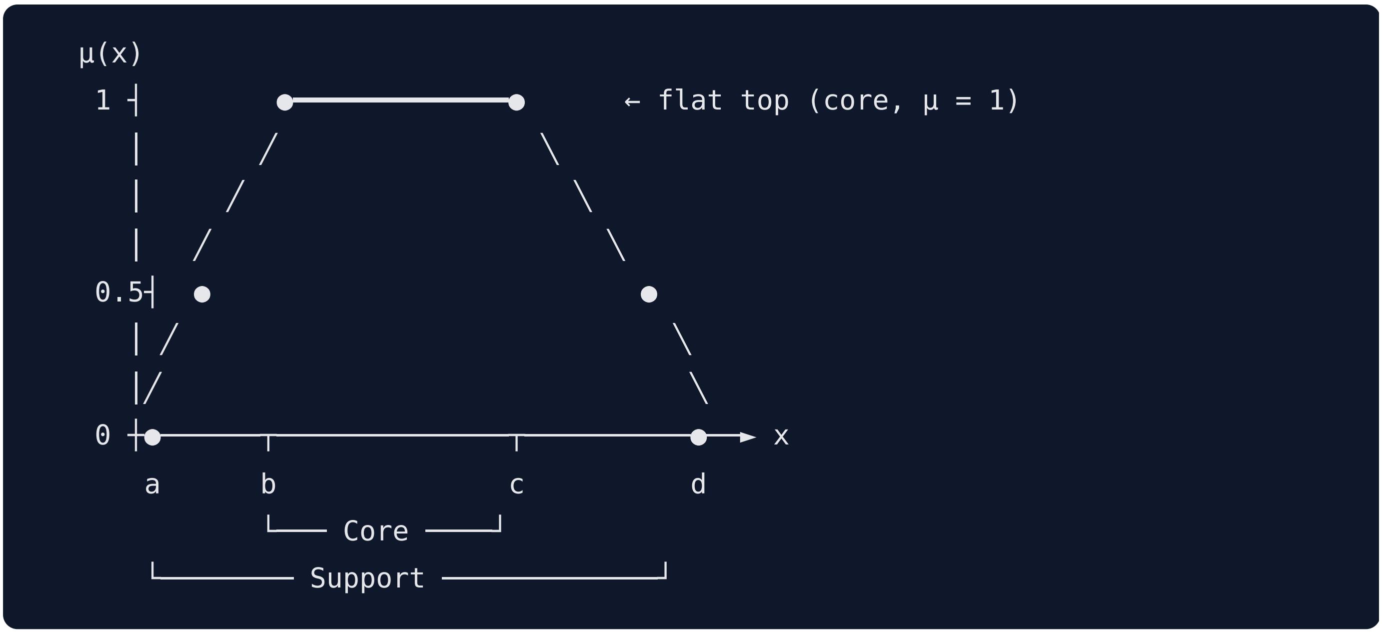
```
 μ(x)
  1 ┤             ●  ← peak at x = b
    │            / \
    │           /   \
    │          /     \
  0.5┤       ●         ●     ← crossover (μ = 0.5)
    │      /           \
    │     /             \
  0 ┼●────┬──────┬──────┬───●──────► x
     a    │      b      │   c
          └─ left slope ──┘
        (rising 1/(b−a))    (falling 1/(c−b))
```

- **Core:** {b}
- **Support:** (a, c)
- **Symmetric** when $b - a = c - b$
- ✅ Simple, fast — most popular MF in fuzzy controllers.

B. Trapezoidal MF

$$\mu(x;a,b,c,d) = \max\left(\min\left(\frac{x-a}{b-a}, 1, \frac{d-x}{d-c}\right), 0\right)$$

Parameters: a = left foot, b = left shoulder, c = right shoulder, d = right foot.

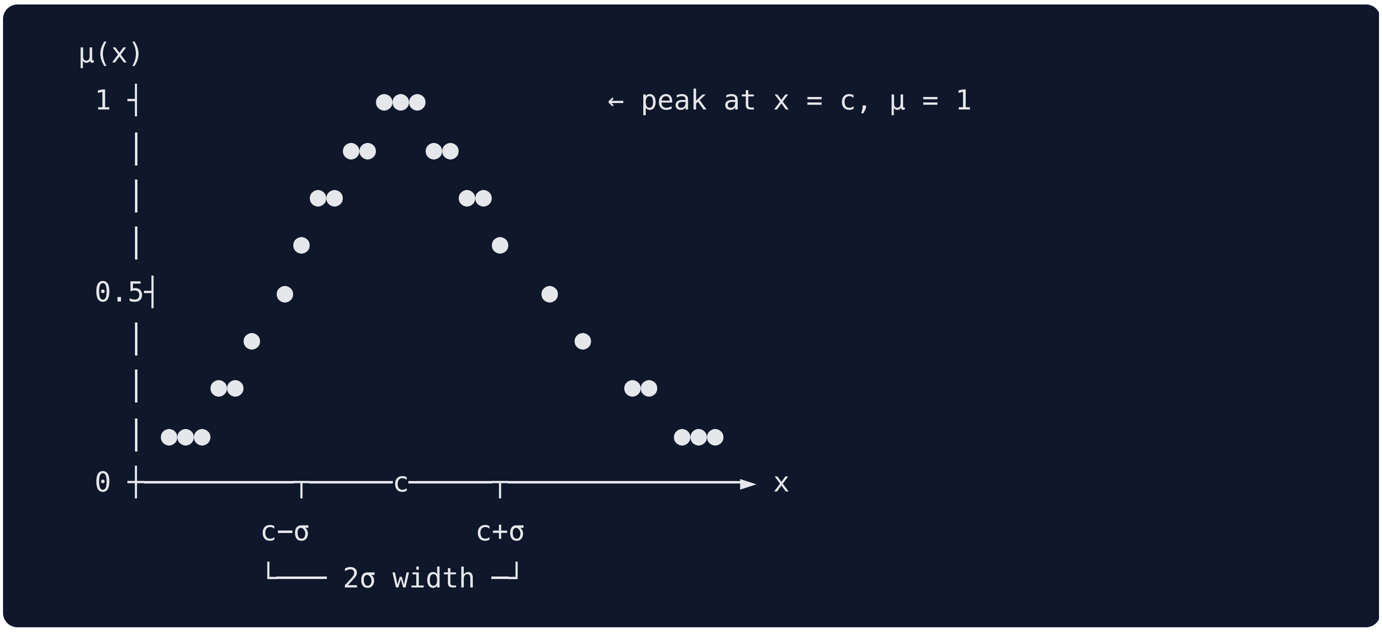


- **Core:** $[b, c]$ (flat plateau)
- **Support:** (a, d)
- ✅ Used when a *range* of values has full membership (e.g., "around 25–30 °C").
- Triangular is a special case where $b = c$.

C. Gaussian MF

$$\mu(x; c, \sigma) = \exp\left(-\frac{(x-c)^2}{2\sigma^2}\right)$$

Parameters: c = center (peak), σ = width (std-dev).



- **Smooth and differentiable** everywhere → ideal for gradient-based tuning (ANFIS).
- **Always > 0** (infinite support).

- Symmetric, single peak.
- Larger $\sigma \Rightarrow$ broader/fuzzier; smaller $\sigma \Rightarrow$ sharper.

D. Generalized Bell MF

$$\mu(x; a, b, c) = \frac{1}{1 + \left|\frac{x - c}{a}\right|^{2b}}$$

Parameters: a = half-width at $\mu = 0.5$, b = slope steepness, c = center.

```
  μ(x)
   1 ┤        ━━━━●━━━━          ← flatter top than Gaussian
     |       /         \
     |      /           \
     |    /               \
   0.5┤   ●                 ●  ← crossover at x = c ± a
     |  /                    \
     | /                       \
     |/                          \
   0 ┼──┬───────c───────┬────────────► x
       c−a             c+a
       └──── 2a width at μ = 0.5 ────┘
```

- 3 parameters $\Rightarrow$ **more flexibility** than Gaussian (controls plateau width independently).
- Larger $b \Rightarrow$ sharper transition (rectangular-like); smaller $b \Rightarrow$ rounder.
- Differentiable — also popular in ANFIS.

E. Sigmoidal MFs (S-shape & Z-shape)

$$S(x; a, c) = \frac{1}{1 + e^{-a(x-c)}}, \qquad Z(x; a, c) = 1 - S(x; a, c)$$

Parameters: a = slope (sign determines direction), c = crossover point ($\mu = 0.5$).

```
  S-shape (open right — "high", "tall", "hot")     Z-shape (open left — "low", "short", "cold")

  μ(x)                                             μ(x)
   1 ┤           ───────────                        1 ┤──────────
     |          /                                     |          \
     |        /                                       |           \
     |      /                                         |             \
   0.5┤    ●                                        0.5┤              ●
     |   /                                            |               \
     |  /                                             |                 \
     |/                                               |                   \
   0 ┼───────c────────────────► x                   0 ┼──────c────────────────► x
          (crossover)                                       (crossover)
```

- **Monotonic** (not bell-shaped) — represents *one-sided* concepts.
- Larger $|a|$ $\Rightarrow$ steeper transition.
- Often used at the *extreme ends* of a variable's term-set (e.g., "very low", "very high").

F. Π (Pi) MF

$$\Pi(x; a, b, c, d) = S(x; a, b) \cdot Z(x; c, d)$$

(Or piecewise polynomial; combines an S-shape rising side with a Z-shape falling side.)

```
  μ(x)
   1 ┤          ╭──────────╮              ← smooth flat top
     |         /            \
     |       /                \
     |     /                    \
   0.5┤    ●                      ●
     |   /                          \
     |  /                             \
     |/                                  \
   0 ┼──┬────┬─────────────┬────┬──────────► x
        a    b             c    d
        └─ rising (S) ─┘   └─ falling (Z) ─┘
```

- **Smooth bell** — generalises trapezoid + Gaussian.
- All parameters tunable: foot of S (a), shoulder (b), shoulder (c), foot of Z (d).
- Continuous & differentiable; good for control surfaces requiring smoothness.

📊 Visual Comparison (Same Universe)

```
  μ(x)                                          ┆
   1 ┤    ┌──────────────┐  ← Trapezoid        ┆
     |    |   ┌───┐      |                      ┆
     |    |  /     \     |     Gaussian         ┆
     |    | /       \    |       .⋅˙⋅.            ┆
     |    |/  /\     \   |     .˙     ˙.   ← Bell ┆
  0.5┤   ╳\ /  \ ╳ ╳     |   .˙        ˙.         ┆
     |   | ╳    \ \ \    |  .˙           ˙.       ┆
     |   |/|    Triangle .˙               ˙.      ┆
     |   / |       \     |.˙                 ˙.   ┆
   0 ┼───┴─┴───────┴─────┴─────────────────────►  x
        a   b         c    d
```

MF type	Params	Smoothness	Speed	Best for
Triangular	3	C^0 (non-smooth)	⚡ fast	Real-time controllers
Trapezoidal	4	C^0	⚡ fast	Range-valued concepts
Gaussian	2	C∞ smooth	medium	ANFIS / gradient tuning
Gen-Bell	3	C∞ smooth	medium	ANFIS with flexible plateau
Sigmoidal	2	C∞ smooth	medium	Open-ended extremes
Π (Pi)	4	C∞ smooth	slow	Smooth controllers

📘 4.3 Fuzzification Methods

Fuzzification = converting a crisp value x_0 (or measurement) into a fuzzy set / membership grade.

A. Singleton fuzzification

$$\mu_{\tilde{A}}(x) = \begin{cases} 1 & x = x_0 \\ 0 & \text{else} \end{cases}$$

Most common in real-time controllers — cheap, exact.

B. Gaussian fuzzification

$$\mu_{\tilde{A}}(x) = \exp\left(-\frac{(x - x_0)^2}{2\sigma^2}\right)$$

Smooth; accounts for measurement noise of std-dev σ.

C. Triangular fuzzification

Spread the crisp value as a narrow triangle around x_0.

D. Intuition / Expert assignment

Domain experts directly draw the MF shape (Saaty's pairwise comparison, Delphi method).

E. Rank ordering / Frequency-based

Use empirical histograms or rank tables to derive μ.

F. Neural Network–based

Train a NN to learn the MF parameters from data (ANFIS uses this).

G. Genetic Algorithm tuning

Parameters of the MF are encoded as chromosomes and evolved.

H. Inductive reasoning / Inference-based

Use entropy minimization (Klir-Yuan) to partition the universe.

✏️ 4.4 Solved Problem — Evaluate an MF

Triangular MF "**young**" $(15, 25, 35)$. Find $\mu(20)$ and $\mu(32)$.

- $\mu(20) = (20 - 15)/(25 - 15) = 5/10 = \mathbf{0.5}$
- $\mu(32) = (35 - 32)/(35 - 25) = 3/10 = \mathbf{0.3}$

Gaussian MF with $c = 25, \sigma = 5$: $\mu(20) = e^{-(5)^2/(2 \cdot 25)} = e^{-0.5} \approx 0.607$.

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5. Fuzzy → Crisp Conversion

📘 5.1 λ-Cuts (Alpha-Cuts) for Fuzzy Sets

The **λ-cut** (also α-cut) of $\tilde{A}$ at level $\lambda \in [0, 1]$:

$$A_\lambda = \{x \in X : \mu_{\tilde{A}}(x) \geq \lambda\}$$

Strong λ-cut: $A_\lambda^+ = \{x : \mu_{\tilde{A}}(x) > \lambda\}$.

📐 Key Properties of λ-Cuts

1. **Nested:** $\lambda_1 < \lambda_2 \Rightarrow A_{\lambda_2} \subseteq A_{\lambda_1}$.

2. **Union:** $(\tilde{A} \cup \tilde{B})_\lambda = A_\lambda \cup B_\lambda$.
3. **Intersection:** $(\tilde{A} \cap \tilde{B})_\lambda = A_\lambda \cap B_\lambda$.
4. **Complement (general):** $(\bar{\tilde{A}})_\lambda \neq \overline{A_\lambda}$ (warning!).
5. **Decomposition (Representation) Theorem:**

$$\tilde{A} = \bigcup_{\lambda \in (0,1]} \lambda \cdot A_\lambda$$

where $\lambda \cdot A_\lambda$ has MF equal to λ on A_λ and 0 elsewhere.

✏️ Example

$\tilde{A} = \frac{0.2}{a} + \frac{0.5}{b} + \frac{0.8}{c} + \frac{1.0}{d} + \frac{0.3}{e}$.

λ	A_λ
0.2	$\{a, b, c, d, e\}$
0.3	$\{b, c, d, e\}$
0.5	$\{b, c, d\}$
0.8	$\{c, d\}$
1.0	$\{d\}$

📘 5.2 λ-Cuts for Fuzzy Relations

If $\tilde{R}$ is a fuzzy relation on $X \times Y$:

$$R_\lambda = \{(x, y) : \mu_{\tilde{R}}(x, y) \geq \lambda\}$$

Yields a crisp relation matrix with entries in $\{0, 1\}$.

✏️ Example

$$M_{\tilde{R}} = \begin{bmatrix} 0.2 & 0.7 \\ 0.9 & 0.4 \end{bmatrix} \xrightarrow{\lambda=0.5} M_{R_{0.5}} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

📘 5.3 Defuzzification Methods

Goal: Convert output fuzzy set $\tilde{C}$ to a crisp value z^*.

A. Centroid / Center of Gravity (COG / COA)

$$z^* = \frac{\int z\mu_{\tilde{C}}(z)dz}{\int \mu_{\tilde{C}}(z)dz} \qquad \text{discrete: } z^* = \frac{\sum_i z_i\mu_{\tilde{C}}(z_i)}{\sum_i \mu_{\tilde{C}}(z_i)}$$

Most popular; balanced.

B. Bisector (BOA)

z^* such that the area is split equally:

$$\int_{z_{\min}}^{z^*} \mu_{\tilde{C}}(z)dz = \int_{z^*}^{z_{\max}} \mu_{\tilde{C}}(z)dz$$

C. Mean of Maxima (MOM)

$$z^* = \frac{\sum_{z\in G} z}{|G|}, \qquad G = \{z : \mu_{\tilde{C}}(z) = \text{height}(\tilde{C})\}$$

D. Smallest / Largest of Maxima (SOM / LOM)

$z^* = \min G$ or $\max G$ respectively.

E. Weighted Average (used in Sugeno)

$$z^* = \frac{\sum_i w_i\bar{z}_i}{\sum_i w_i}$$

where $\bar{z}_i$ is the peak of the i-th output MF and w_i its firing strength.

F. Center of Sums (COS)

Faster than centroid; sums areas without union:

$$z^* = \frac{\sum_z z \sum_i \mu_{\tilde{C}_i}(z)}{\sum_z \sum_i \mu_{\tilde{C}_i}(z)}$$

Comparison

Method	Pros	Cons
Centroid	Smooth, intuitive	Computationally heavy
Bisector	Balances area	Same as centroid for symmetric
MOM	Simple, fast	Ignores shape
SOM/LOM	Conservative / aggressive choice	Discontinuous
Weighted Avg	Very fast	Only for Sugeno-style outputs

✏️ 5.4 Solved Problem — Defuzzification

Output fuzzy set $\tilde{C}$ on $Z = \{0, 1, 2, 3, 4, 5, 6\}$ with MF $\mu = (0, 0.3, 0.6, 1.0, 1.0, 0.5, 0.2)$.

(a) Centroid

$$z^* = \frac{0(0) + 1(0.3) + 2(0.6) + 3(1) + 4(1) + 5(0.5) + 6(0.2)}{0 + 0.3 + 0.6 + 1 + 1 + 0.5 + 0.2} = \frac{0 + 0.3 + 1.2 + 3 + 4 + 2.5 + 1.2}{3.6} = \frac{12.2}{3.6} \approx \mathbf{3.39}$$

(b) MOM

Maxima occur at $z = 3, 4$: $z^* = (3 + 4)/2 = \mathbf{3.5}$.

(c) SOM / LOM

SOM = 3, LOM = 4.

(d) Bisector

Total area = 3.6. Half = 1.8. Cumulative:
$0, 0.3, 0.9, 1.9, 2.9, 3.4, 3.6 \rightarrow$ crosses 1.8 between $z = 2$ (0.9) and $z = 3$ (1.9). Linearly: $z^* \approx 2 + (1.8 - 0.9)/(1.9 - 0.9) \approx 2.9$.

---

6. Classical vs Fuzzy Logic

📘 6.1 Classical Predicate Logic

Propositions are binary — *true* (1) or *false* (0).

Connective	Symbol	Definition
Negation	$\neg p$	$1 - p$
Conjunction	$p \wedge q$	$\min$ / AND
Disjunction	$p \vee q$	$\max$ / OR
Implication	$p \rightarrow q$	$\neg p \vee q$
Bi-conditional	$p \leftrightarrow q$	$(p \rightarrow q) \wedge (q \rightarrow p)$

Predicate logic adds quantifiers $\forall$ (universal) and $\exists$ (existential).

Classical Inference Rules

1. **Modus Ponens (MP):** $\frac{p, p \rightarrow q}{q}$
2. **Modus Tollens (MT):** $\frac{\neg q, p \rightarrow q}{\neg p}$
3. **Hypothetical syllogism:** $\frac{p \rightarrow q, q \rightarrow r}{p \rightarrow r}$

📘 6.2 Fuzzy Logic

Fuzzy logic generalizes Boolean logic to truth values in $[0, 1]$. Connectives are realized via t-norms (AND), s-norms (OR), and negation operators.

Operator	Standard form	Probabilistic	Łukasiewicz
NOT p	$1 - p$	$1 - p$	$1 - p$
$p \wedge q$	$\min$	$p \cdot q$	$\max(0, p + q - 1)$
$p \vee q$	$\max$	$p + q - pq$	$\min(1, p + q)$

Linguistic variables (e.g., "temperature") take **linguistic values** ("low", "medium", "high"), each represented by a fuzzy set.

7. Approximate Reasoning & Fuzzy Implication

📘 7.1 Approximate Reasoning

Reasoning where premises, rules, and conclusions involve fuzzy concepts.

Generalized Modus Ponens (GMP):

- **Premise 1 (rule):** IF x is $\tilde{A}$ THEN y is $\tilde{B}$
- **Premise 2 (fact):** x is $\tilde{A}'$
- **Conclusion:** y is $\tilde{B}'$

Computed via the **compositional rule of inference (CRI)** by Zadeh:

$$\tilde{B}' = \tilde{A}' \circ \tilde{R}(\tilde{A}, \tilde{B})$$

where $\tilde{R}$ is the fuzzy implication relation.

🧮 7.2 Fuzzy Implication Operators

Different definitions for $\mu_{\tilde{R}}(x, y) = \mathrm{Imp}(\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(y))$:

Name	Formula
Mamdani (Minimum)	$\min(\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(y))$

Larsen (Product)	$\mu_{\tilde{A}}(x) \cdot \mu_{\tilde{B}}(y)$
Zadeh	$\max(\min(\mu_{\tilde{A}}, \mu_{\tilde{B}}), 1 - \mu_{\tilde{A}})$
Łukasiewicz	$\min(1, 1 - \mu_{\tilde{A}} + \mu_{\tilde{B}})$
Gödel	1 if $\mu_{\tilde{A}} \le \mu_{\tilde{B}}$ else $\mu_{\tilde{B}}$
Kleene-Dienes	$\max(1 - \mu_{\tilde{A}}, \mu_{\tilde{B}})$
Goguen	1 if $\mu_{\tilde{A}} \le \mu_{\tilde{B}}$ else $\mu_{\tilde{B}}/\mu_{\tilde{A}}$

> **Mamdani** and **Larsen** are popular in **engineering** controllers (not classical logic implications, but work well in practice).

✏️ 7.3 Solved Problem — GMP with Mamdani Implication

Universes $X = \{x_1, x_2\}$, $Y = \{y_1, y_2, y_3\}$.

Rule: IF x is $\tilde{A}$ THEN y is $\tilde{B}$, with $\tilde{A} = \frac{0.6}{x_1} + \frac{0.9}{x_2}$, $\tilde{B} = \frac{0.4}{y_1} + \frac{0.7}{y_2} + \frac{1.0}{y_3}$.

Step 1 — Build relation $\tilde{R}$ (Mamdani = min):

$$\tilde{R} = \begin{bmatrix} \min(0.6, 0.4) & \min(0.6, 0.7) & \min(0.6, 1.0) \\ \min(0.9, 0.4) & \min(0.9, 0.7) & \min(0.9, 1.0) \end{bmatrix} = \begin{bmatrix} 0.4 & 0.6 & 0.6 \\ 0.4 & 0.7 & 0.9 \end{bmatrix}$$

Step 2 — Fact: $\tilde{A}' = \frac{0.5}{x_1} + \frac{0.8}{x_2}$.

Step 3 — Inference (max-min):

$$\mu_{\tilde{B}'}(y_1) = \max(\min(0.5, 0.4), \min(0.8, 0.4)) = \max(0.4, 0.4) = 0.4$$

$$\mu_{\tilde{B}'}(y_2) = \max(\min(0.5, 0.6), \min(0.8, 0.7)) = \max(0.5, 0.7) = 0.7$$

$$\mu_{\tilde{B}'}(y_3) = \max(\min(0.5, 0.6), \min(0.8, 0.9)) = \max(0.5, 0.8) = 0.8$$

Conclusion: $\tilde{B}' = \frac{0.4}{y_1} + \frac{0.7}{y_2} + \frac{0.8}{y_3}$.

---

8. Fuzzy Rule-Based Systems & Inference

📘 8.1 Linguistic Variables & Hedges

A **linguistic variable** $\langle$ name, $T(\text{name}), U, G, M\rangle$:

- name e.g. *Age*
- term set T = {young, old, very-young, somewhat-old…}

- universe $U = [0, 120]$
- syntactic rule G for generating terms
- semantic rule M assigning MFs

Linguistic Hedges (Modifiers)

Given fuzzy set $\tilde{A}$ with MF $\mu_{\tilde{A}}$, hedges produce new fuzzy sets:

Hedge	Operation	Effect
Very A	$[\mu_{\tilde{A}}(x)]^2$	Concentration — sharper
More-or-less A	$[\mu_{\tilde{A}}(x)]^{0.5}$	Dilation — softer
Extremely A	$[\mu_{\tilde{A}}(x)]^3$	Strong concentration
Slightly A	$[\mu_{\tilde{A}}(x)]^{0.33}$	Strong dilation
Indeed A	$\begin{cases} 2\mu^2 & \mu \leq 0.5 \\ 1 - 2(1-\mu)^2 & \mu > 0.5 \end{cases}$	Contrast intensification
Not A	$1 - \mu_{\tilde{A}}(x)$	Negation
About A	bell around peak	Approximation

✏️ Example

"Tall" $= \frac{0.0}{150} + \frac{0.5}{170} + \frac{1.0}{180}$
"Very Tall" $= \frac{0.0}{150} + \frac{0.25}{170} + \frac{1.0}{180}$
"More-or-less Tall" $= \frac{0.0}{150} + \frac{0.707}{170} + \frac{1.0}{180}$

📘 8.2 Fuzzy Rule Base

A **fuzzy rule** has the form:

$$\text{IF } x_1 \text{ is } \tilde{A}_1 \text{ AND } x_2 \text{ is } \tilde{A}_2 \text{ THEN } y \text{ is } \tilde{B}$$

Multiple rules form a **rule base** $\{R_i\}_{i=1}^N$.

Connectives within antecedent

- `AND` $\rightarrow$ t-norm (often min or product)
- `OR` $\rightarrow$ s-norm (often max)

Aggregation of Rules

Once each rule fires to produce $\tilde{B}_i'$, all rule outputs are **combined**:

$$\tilde{B}' = \bigcup_{i=1}^{N} \tilde{B}'_i$$

Typical aggregation operators:

Method	Definition
Max	$\mu_{\tilde{B}'}(y) = \max_i \mu_{\tilde{B}'_i}(y)$ — most common
Sum	$\sum_i \mu_{\tilde{B}'_i}(y)$
Probabilistic OR	iteratively $a + b - ab$

📘 8.3 Fuzzy Inference System (FIS)

```
Crisp input ──► Fuzzification ──► Rule Evaluation (Implication)
                                          │
                                          ▼
                                     Aggregation
                                          │
                                          ▼
                                 Defuzzification ──► Crisp output
```

📘 8.4 Mamdani Fuzzy Model (1975)

- **Antecedent:** fuzzy MFs
- **Consequent:** fuzzy MFs
- **Implication:** min (clipping) or product (scaling)
- **Aggregation:** max
- **Defuzzification:** Centroid / MOM / etc.

Steps

1. Fuzzify each crisp input $x_0 \to \mu_{\tilde{A}_i}(x_0)$.
2. Compute rule firing strength $w_i = T(\mu_{\tilde{A}_{i1}}(x_1^0), \ldots)$.
3. **Implication**: clip/scale consequent MF by $w_i \to \tilde{B}'_i$.
4. **Aggregate** all $\tilde{B}'_i \to \tilde{B}'$.
5. **Defuzzify** $\tilde{B}' \to$ crisp output y^*.

Pros & Cons

✅ Intuitive, expressive consequents — easy to interpret.
❌ Computationally expensive (defuzzification on full fuzzy set).

📘 8.5 Sugeno (TSK) Fuzzy Model (Takagi-Sugeno-Kang, 1985)

- **Antecedent:** fuzzy
- **Consequent:** crisp **function** of inputs:

$$\text{IF } x_1 \text{ is } \tilde{A}_1 \text{ AND } x_2 \text{ is } \tilde{A}_2 \text{ THEN } y_i = f_i(x_1, x_2)$$

- **Zero-order Sugeno:** $f_i = c_i$ (constant)
- **First-order Sugeno:** $f_i = a_{i0} + a_{i1}x_1 + a_{i2}x_2$

Crisp output (weighted average)

$$y^* = \frac{\sum_{i=1}^{N} w_i \cdot f_i(\mathbf{x})}{\sum_{i=1}^{N} w_i}$$

Pros & Cons

✅ Computationally efficient, no defuzzification step.
✅ Works seamlessly with adaptive techniques (ANFIS).
❌ Consequents less interpretable than Mamdani.

Mamdani vs Sugeno

Aspect	Mamdani	Sugeno
Consequent	Fuzzy MF	Linear / constant function
Defuzzification	Centroid etc.	Weighted average
Interpretability	High	Medium
Speed	Slow	Fast
Use case	Decision support, controllers requiring intuitive rules	Adaptive control, optimization

---

✏️ 8.6 Solved Problem — Complete Mamdani FIS

Problem: Tip calculator. Inputs: *service* $(0\text{–}10)$, *food* $(0\text{–}10)$. Output: *tip* $(0\text{–}25\%)$.

MFs

- service: poor `(0,0,5)`, good `(0,5,10)`, excellent `(5,10,10)` (triangular)
- food: bad `(0,0,5)`, good `(5,10,10)`
- tip: low `(0,5,10)`, average `(10,15,20)`, high `(20,25,25)`

Rules

- R1: IF service is poor OR food is bad THEN tip is low
- R2: IF service is good THEN tip is average
- R3: IF service is excellent OR food is good THEN tip is high

Input: service = 3, food = 8.

Step 1 — Fuzzification

Variable	poor	good	excellent	bad	good
service=3	$(5-3)/5 = 0.4$	$3/5 = 0.6$	0	—	—
food=8	—	—	—	0	$(8-5)/5 = 0.6$

Step 2 — Rule firing (OR=max, AND=min)

- $w_1 = \max(0.4, 0) = 0.4$ → tip = low
- $w_2 = 0.6$ → tip = average
- $w_3 = \max(0, 0.6) = 0.6$ → tip = high

Step 3 — Implication (clip consequent)

- low_clip = min(0.4, low)
- average_clip = min(0.6, average)
- high_clip = min(0.6, high)

Step 4 — Aggregation: max of the three clipped MFs ⇒ output fuzzy set $\tilde{B}'$.

Step 5 — Defuzzification (centroid). Approximate by sampling tip ∈ {0,5,10,15,20,25} and computing z^*. For symmetric triangles the centroid ≈ weighted average of peaks:

$$y^* \approx \frac{0.4(5) + 0.6(15) + 0.6(25)}{0.4 + 0.6 + 0.6} = \frac{2 + 9 + 15}{1.6} = \frac{26}{1.6} \approx \mathbf{16.25\%}$$

✏️ 8.7 Solved Problem — Zero-Order Sugeno

Same inputs, rules:

- R1: tip = 5
- R2: tip = 15

- R3: tip = 25

$$y^* = \frac{0.4(5) + 0.6(15) + 0.6(25)}{0.4 + 0.6 + 0.6} = \frac{26}{1.6} = \mathbf{16.25\%}$$

(In this case Mamdani-with-symmetric-peaks ≈ Sugeno result.)

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9. Applications of Fuzzy Logic

📘 9.1 Home Appliances

Washing Machine

- **Inputs:** dirt level (turbidity sensor), fabric type, load weight.
- **Outputs:** wash time, water level, agitation cycle, detergent quantity.
- **Sample rule:** IF dirt is HIGH AND fabric is HARD THEN wash-time is LONG.

Air Conditioner

- **Inputs:** room temperature, humidity, occupancy.
- **Outputs:** compressor speed, fan speed.
- **Sample rule:** IF temp is HOT AND humidity is HIGH THEN cooling is STRONG.

Rice Cooker, Refrigerator, Microwave, Vacuum cleaner

- Similar fuzzy controllers; Japan ("Sendai Subway", 1987) is the **classic large-scale industrial example**.

📘 9.2 General Fuzzy Logic Controller (FLC)

```
                    ┌─────────────────────┐
                    │     Plant /         │
    r(t) ─┐        │     Process         │
          │        └─────────────────────┘
          ▼                  │ y(t)
       ┌───┐    e(t)    ┌──────────┐     crisp
       │ + │──────────► │  Fuzzy   │ ◄───────────┐
       └───┘            │ Logic    │     sensors│
          ▲             │ Control. │             │
          │             └──────────┘             │
          │ y(t)             │ u(t)              │
          └──────────────────┴───────────────────┘
```

- **Inputs:** error $e = r - y$, rate $\dot{e}$.
- **MFs:** NB, NM, NS, ZE, PS, PM, PB (negative big, … positive big).
- **Rule table (FAM — Fuzzy Associative Memory):**

$\dot{e} \backslash e$	**NB**	**NM**	**NS**	**ZE**	**PS**	**PM**	**PB**
NB	NB	NB	NB	NM	NS	ZE	PS
NM	NB	NB	NM	NS	ZE	PS	PM
NS	NB	NM	NS	ZE	PS	PM	PB
ZE	NM	NS	ZE	ZE	ZE	PS	PM
PS	NM	NS	ZE	PS	PS	PM	PB
PM	NM	NS	ZE	PS	PM	PB	PB
PB	NS	ZE	PS	PM	PB	PB	PB

Output u = controller signal (e.g., motor voltage).

📘 9.3 Medical Diagnosis (Fuzzy Inference)

Approach

- Symptoms $S = \{s_1, \dots, s_m\}$, diseases $D = \{d_1, \dots, d_n\}$.
- Fuzzy relation $\tilde{R}$ on $S \times D$: $\mu_{\tilde{R}}(s, d) =$ how strongly s indicates d.
- Patient profile $\tilde{P}$ on S.
- Diagnosis: $\tilde{D} = \tilde{P} \circ \tilde{R}$ (max-min).

✏️ Example

$S = \{\text{fever, cough, fatigue}\}$, $D = \{\text{flu, cold}\}$.

$$\tilde{R} = \begin{bmatrix} 0.9 & 0.4 \\ 0.6 & 0.8 \\ 0.7 & 0.3 \end{bmatrix}, \quad \tilde{P} = (0.8, 0.5, 0.6)$$

$\mu_{\tilde{D}}(\text{flu}) = \max(\min(0.8, 0.9), \min(0.5, 0.6), \min(0.6, 0.7)) = \max(0.8, 0.5, 0.6) = \mathbf{0.8}$
$\mu_{\tilde{D}}(\text{cold}) = \max(\min(0.8, 0.4), \min(0.5, 0.8), \min(0.6, 0.3)) = \max(0.4, 0.5, 0.3) = \mathbf{0.5}$

⇒ Flu is the more likely diagnosis.

📘 9.4 Weather Forecasting

- **Inputs:** temperature, humidity, pressure trend, wind speed, cloud cover.
- **Linguistic terms:** low / medium / high.
- **Sample rules:**
  - IF pressure is FALLING AND humidity is HIGH THEN rain-chance is HIGH.
  - IF temp is HIGH AND humidity is LOW AND wind is LOW THEN weather is SUNNY.
- **Output:** probability of rain, expected temperature range — defuzzified into crisp forecast.

Advantages over Bayesian-only forecasting: copes with linguistic expert knowledge, handles missing/noisy data.

---

10. Quick Revision Cheat-Sheet

📝 Classical vs Fuzzy

- Crisp: $\chi_A \in \{0,1\}$. Fuzzy: $\mu_{\tilde{A}} \in [0,1]$.
- All classical laws hold for fuzzy EXCEPT **Excluded Middle** & **Contradiction**.

📝 Operations

- $\cup = \max, \cap = \min, ^{-} = 1 - \mu$ (Zadeh standard).
- Generalized: t-norm (AND), s-norm (OR).
- Algebraic: product $a \cdot b$; algebraic sum $a + b - ab$.

📝 Fuzzy Relations

- $\mu_{\tilde{R}} : X \times Y \to [0,1]$ — matrix form.
- **Max-min composition**: $\max_y \min(\mu_R, \mu_S)$.
- Properties: reflexive, symmetric, transitive ⇒ fuzzy equivalence.

📝 MF & Fuzzification

- Features: core, support, boundary, height, crossover.
- Convex iff every α-cut is convex (theorem).
- Standard: triangular, trapezoidal, Gaussian, bell, sigmoid.
- Methods: singleton, Gaussian, expert, NN, GA, inductive.

📝 λ-Cut & Defuzzification

- $A_\lambda = \{x : \mu(x) \geq \lambda\}$ — nested.
- Decomposition: $\tilde{A} = \bigcup_\lambda \lambda \cdot A_\lambda$.

- Centroid $z^* = \frac{\sum z\mu}{\sum \mu}$ — most common.
- MOM, SOM, LOM, Bisector, Weighted-Avg (Sugeno).

📝 Implication & GMP

- Mamdani = min, Larsen = product.
- GMP: $\tilde{B}' = \tilde{A}' \circ \tilde{R}$.

📝 Linguistic Hedges

- very = μ^2, more-or-less = $\mu^{0.5}$, extremely = μ^3.

📝 Mamdani vs Sugeno

	Mamdani	Sugeno
Output	Fuzzy MF	Crisp function
Defuzz	Centroid	Weighted avg
Speed	Slow	Fast
Use	Interpretable rules	Adaptive control / ANFIS

📝 Applications

- Washing machine, AC, rice cooker, vacuum, ABS brakes.
- Medical diagnosis via max-min composition $\tilde{D} = \tilde{P} \circ \tilde{R}$.
- Weather forecasting (rules from meteorologists).
- General FLC with FAM table over $(e, \dot{e})$.

> **End of Unit 2 — Fuzzy Sets, Logic & Systems.**
> *Next: Unit 3 (share when ready — likely Neural Networks in depth or Genetic Algorithms in depth).*